

# 20 Questions and Answers on Retrieval-Augmented Generation (RAG)

## **Q1. What is Retrieval-Augmented Generation (RAG)?**

A1. RAG is an AI architecture that combines retrieval of external knowledge from a database or document store with generative language models to produce accurate, context-aware responses.

## **Q2. Why do we need RAG when we already have LLMs?**

A2. LLMs have a knowledge cutoff and may hallucinate. RAG allows them to fetch up-to-date, domain-specific knowledge, improving accuracy and reliability.

## **Q3. What are the two main components of RAG?**

A3. The retriever (fetches relevant documents from a knowledge base) and the generator (an LLM that generates answers using retrieved context).

## **Q4. Which vector databases are commonly used in RAG?**

A4. Popular choices include Pinecone, Weaviate, FAISS, Milvus, and Qdrant.

## **Q5. What role does embedding play in RAG?**

A5. Embeddings convert text into numerical vectors so that semantic similarity search can retrieve the most relevant documents.

## **Q6. How does RAG differ from fine-tuning an LLM?**

A6. Fine-tuning modifies model weights with new data, while RAG augments inference dynamically without retraining by adding external context.

## **Q7. What is chunking in RAG?**

A7. Chunking is splitting large documents into smaller pieces so embeddings can represent them effectively for retrieval.

## **Q8. How does RAG reduce hallucinations?**

A8. By grounding answers in retrieved external data instead of relying solely on the model's internal memory.

## **Q9. What is the difference between dense retrieval and sparse retrieval?**

A9. Dense retrieval uses embeddings for semantic similarity, while sparse retrieval (like BM25) relies on keyword matching.

## **Q10. Which embedding models are commonly used for RAG?**

A10. OpenAI's text-embedding-ada-002, Sentence-BERT, Cohere embeddings, and Instructor models are popular.

**Q11. What is hybrid retrieval in RAG?**

A11. It combines dense and sparse retrieval to improve both recall and precision.

**Q12. How do you evaluate a RAG system?**

A12. Metrics include precision@k, recall@k for retrieval, and answer relevance, factual correctness, and hallucination rate for generation.

**Q13. What challenges exist in building RAG?**

A13. Challenges include choosing the right chunk size, latency issues, irrelevant retrieval, and handling rapidly changing knowledge bases.

**Q14. How do you handle multi-turn conversations in RAG?**

A14. By maintaining conversation history and passing it as context along with retrieved knowledge.

**Q15. Can RAG work in domain-specific industries like healthcare or law?**

A15. Yes, by indexing domain-specific documents and ensuring compliance with data privacy and regulations.

**Q16. What is retrieval latency, and how do you optimize it?**

A16. Retrieval latency is the delay in fetching documents. Optimization methods include efficient vector DBs, approximate nearest neighbor (ANN) search, and caching.

**Q17. How does caching improve RAG performance?**

A17. Frequently asked queries and their results can be cached to avoid repeated retrieval and reduce costs.

**Q18. What is the difference between open-domain and closed-domain RAG?**

A18. Open-domain retrieves from large, general knowledge bases (like Wikipedia), while closed-domain uses specific enterprise or industry datasets.

**Q19. What is the role of prompt engineering in RAG?**

A19. Prompt engineering ensures that retrieved knowledge is effectively integrated into the LLM's response generation.

**Q20. How does RAG compare with Agentic AI systems?**

A20. RAG focuses on retrieval + generation, while Agentic AI adds reasoning, planning, and tool-use capabilities on top of RAG principles.